

Math 421

Wednesday, December 10

Announcements

- This week's drop-in hours –
 - **Today** 12:00-1:00 pm VV B205
 - Thursday 9:00 – 11:00 am VV B205
- Final Exam –
 - Date: Monday, December 15th
 - Time: 5:05-7:05 pm
 - Location: Bardeen 140
- Course Evaluations!
 - Open until 11:59pm **today!**

Continuity $\Rightarrow$ Integrability

Theorem 33.2. Every continuous function f on $[a, b]$ is integrable.

Proof. Let $\epsilon > 0$. Since f is continuous on $[a, b]$, it is uniformly continuous on every closed subinterval, so there exists

$$\delta > 0 \text{ such that for all } x, y \in [a, b] \text{ if } |x - y| < \delta \text{ then } |f(x) - f(y)| < \boxed{}.$$

Let P be a partition of $[a, b]$ such that $t_k - t_{k-1} < \delta$. By the Extreme Value Theorem, on each $[t_k, t_{k-1}]$ there exists x_k and y_k such that $M_k = f(x_k) = \sup\{f(x): x \in [t_{k-1}, t_k]\}$ and $m_k = f(y_k) = \inf\{f(x): x \in [t_{k-1}, t_k]\}$. Then

$$U(f, P) - L(f, P) = \sum_{k=1}^n (M_k - m_k)(t_k - t_{k-1})$$

$$= \sum_{k=1}^n (f(x_k) - f(y_k))(t_k - t_{k-1})$$

$$< \sum_{k=1}^n (t_k - t_{k-1})$$

$$\square (b - a) < \epsilon$$

Fundamental Theorem of Calculus I

Theorem 24.1. If g is a continuous function on $[a, b]$ that is differentiable on (a, b) , and if g' is integrable on $[a, b]$, then

$$\int_a^b g' = g(b) - g(a)$$

Proof

Proof (cont.)

Fundamental Theorem of Calculus II

Theorem 34.3. Let f be an integrable function on $[a, b]$. For $x \in [a, b]$, let

$$F(x) = \int_a^x f(t) dt$$

Then F is continuous on $[a, b]$. If f is continuous at x_0 in (a, b) , then F is differentiable at x_0 and

$$F'(x_0) = f(x_0)$$

Proof

Proof

Proof (cont.)

Integration by Parts

Theorem 34.2. If u and v are continuous functions on $[a, b]$ that are differentiable on (a, b) , and if u' and v' are integrable on $[a, b]$, then

$$\int_a^b u(x)v'(x) dx + \int_a^b u'(x)v(x) dx = u(b)v(b) - u(a)v(a)$$

Activity: How does this follow from FTC 1? What should g be?

Change of Variable (U – Substitution)

Theorem 34.4. Let u be a differentiable function on an open interval J such that u' is continuous, and let I be an open interval such that $u(x) \in I$ for all $x \in J$. if f is continuous on I , then $f \circ u$ is continuous on J and

$$\int_a^b f \circ u(x) u'(x) dx = \int_{u(a)}^{u(b)} f(u) du$$

For a, b in J .